Direct-root Cartan Schur, sequential secants, and rational conditional trace reduction

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

R-084 reduced controlled Cartan CFAR exactly to a root-diagonal Ornstein–Uhlenbeck gradient square. R-085 then proved a valid nonorthogonal Schur theorem for a stronger expression carrying an additional outer factor 2^j. R-087 proved its spatial atom estimate but not the one-use ledger. A formula-level audit now shows that the exact R-084 target does not carry that outer factor.

This note makes four advances.

1. The exact R-084 target obeys a direct-root nonorthogonal Schur theorem for every $s > 0$, with the unweighted ledger $\sum_k q_k$. At $s = 7/12$ the balanced constant is
$$16.30295538482827\ldots$$
and the far-gap factor remains $2^{-7C/6}$.

2. A sequential physical-shell secant gives an exact three-channel Cartan atom. Future control shells are absent from the coefficient path of the kth atom, and its pointwise envelope has no artificial sixth power of the complete translated model norm.

3. For $0 < s < 1$, the deterministic quartic payload satisfies
$$\|A^{\otimes 3} \otimes DA\|_{B^s_{1,1}} \lesssim_s \|A\|_{H^2}\|A\|_6^3.$$
Its right side is exactly the critical pure-control product $X_A^{1/2}Y_A^{1/2}$.

4. The rational K_η square, transformed Wick term, and η-trace debt obey an exact pointwise three-term null identity. A conditional moment theorem identifies precisely when retained-square variance cancels the debt and precisely which mean and covariance defects survive.

These results do not prove the production sequential secant-to-quartic estimate, direct integrated Cartan CFAR, the coefficient-dominant rational causal packet, REG, uniform OVERLAP, controlled-shell one-use, Nelson, an interacting measure, or Sector-A closure. Tier stays T4.

The load-bearing internal authorities are R-050, R-066, R-074, R-077, R-079, R-081, and R-083–R-087. A targeted search of the external `Contents` archive found no relevant Cartan-current, OU-gradient, paracomposition, or rational trace theorem; its Cartan material concerns an unrelated Lie-algebra problem.

2 Exact target and normalization audit

Write

$$\mathrm{Q}_{j,C} f = (\Pi_m f)_{m \geq j+C}. \tag{2.1}$$

After separating the accepted finite-low object, absorb the OU action into the atoms by defining

$$v_{A,j,k,t} := P_t^{(j)} u_{A,j,k,t}. \tag{2.2a}$$

The linearity of $P_t^{(j)}$ then puts the exact R-084 target in the form

$$\boxed{\mathcal{S}_C = \sum_{A,j} \int_0^\infty e^{-2t} \mathbb{E} \left\| \mathrm{Q}_{j,C} \sum_{k \leq j} v_{A,j,k,t} \right\|_{\mathrm{HS}}^2 dt.} \tag{2.2}$$

The controlled Cartan coefficient is

$$S_C^{\mathrm{ctrl}} = \frac{3}{40P} \mathcal{S}_C. \tag{2.3}$$

There is no factor 2^j outside the root sum in (2.2).

By contrast, R-085 Theorem 4.1 proves a correct theorem for

$$\sum_m \sum_{j \leq m-C} 2^j \mathbb{E} \|\Pi_m f_j\|^2. \tag{2.4}$$

The audit therefore narrows the sufficient architecture; it does not withdraw the stronger R-085 theorem. If a hidden rescaling $u_{j,k} = 2^{-j/2} \widetilde{u}_{j,k}$ is intended, it must be stated and checked against the unscaled atoms in R-087. No such rescaling occurs in the current formulas.

This fires `AUDIT-2026-07-25-A13-R085-CARTAN-OUTER-WEIGHT-NORMALIZATION`. The correction changes the sufficient ledger and threshold, not any already-proved R-085 inequality.

3 Direct-root nonorthogonal Schur theorem

Theorem 3.1

Let $v_{A,j,k,t}$ be Hilbert-valued atoms for $k \leq j$, and suppose that for $m \geq j + C_0$,

$$\sum_A \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m v_{A,j,k,t}\|_{\mathrm{HS}}^2 dt \leq 2^{-2s(m-k)} q_k \tag{3.1}$$

with $s > 0$ and deterministic $q_k \geq 0$. For every $C \geq C_0$ and $0 < \eta < 2s$,

$$\boxed{\begin{aligned} &\sum_{A,j} \int_0^\infty e^{-2t} \mathbb{E} \left\| \mathrm{Q}_{j,C} \sum_{k \leq j} v_{A,j,k,t} \right\|_{\mathrm{HS}}^2 dt \\ &\qquad \leq \frac{2^{-2sC}}{(1-2^{-\eta})(1-2^{-2s})(1-2^{\eta-2s})} \sum_k q_k. \end{aligned}} \tag{3.2}$$

No orthogonality between different k is assumed.

Proof. For fixed A, j, m, t, weighted Cauchy–Schwarz gives

$$\left\| \sum_{k \leq j} \Pi_m v_{A,j,k,t} \right\|^2 \leq \frac{1}{1-2^{-\eta}} \sum_{k \leq j} 2^{\eta(j-k)} \|\Pi_m v_{A,j,k,t}\|^2. \tag{3.3}$$

Insert (3.1), exchange nonnegative sums, and put $r = j - k$. Then

$$\sum_{m \geq j+C} 2^{-2s(m-k)} = \frac{2^{-2s(C+r)}}{1-2^{-2s}}, \tag{3.4}$$

while

$$\sum_{r \geq 0} 2^{(\eta-2s)r} = \frac{1}{1-2^{\eta-2s}}. \tag{3.5}$$

The latter converges exactly when $\eta < 2s$. Combining (3.3)–(3.5) proves (3.2). □

Balanced value and sharp threshold

For fixed s, the two η-dependent factors are balanced at $\eta = s$. Indeed, if $a = 2^{-\eta}$ and $b = 2^{-(2s-\eta)}$, then $ab = 2^{-2s}$ and $(1-a)(1-b)$ is maximal at $a = b$. Hence

$$C_s^{\mathrm{dir}} = \frac{1}{(1-2^{-s})^2(1-2^{-2s})}. \tag{3.6}$$

At $s = 7/12$,

$$C_{7/12}^{\mathrm{dir}} = 16.30295538482827\ldots, \qquad 2^{-2sC} = 2^{-7C/6}. \tag{3.7}$$

The threshold $s > 0$ is sharp under only (3.1). At $s = 0$, one fixed j, k may have a unit component in every output shell $m \geq j + C$. The atom hypothesis still holds with $q_k = 1$, but the squared ℓ_m^2 norm grows linearly with the output truncation. Thus no cutoff-uniform conclusion follows at $s = 0$.

Conditional Cartan corollary

If the production atoms satisfy (3.1) and

$$\boxed{\sum_k q_k \leq K_e \left\{1 + \mathbb{E}[X^{1/2}Y^{1/2}]\right\},} \tag{3.8}$$

then

$$S_C^{\mathrm{ctrl}} \leq \frac{3K_e}{40P} C_{s,\eta}^{\mathrm{dir}} 2^{-2sC} \left\{1 + \mathbb{E}[X^{1/2}Y^{1/2}]\right\}. \tag{3.9}$$

Equation (3.8), not the Schur summation, remains open. The direct-root sufficient threshold is $s > 0$, not $s > 1/2$.

4 Exact sequential Cartan secant

Fix the generator and root scale and write

$$\Phi = \Phi_{A,j}, \qquad \mathcal{C}_\Phi(z) = \Phi(z)^T Dz. \tag{4.1}$$

Let

$$A^{(k)} = A^{(\ell)} + \sum_{r=j_0}^{k} a_r, \qquad B_{j,k} = w_j + A^{(k-1)}. \tag{4.2}$$

Define

$$\Theta_{A,j,k} = \mathcal{C}_\Phi(B_{j,k} + a_k) - \mathcal{C}_\Phi(B_{j,k}). \tag{4.3}$$

Then the physical-shell telescope is exact:

$$\sum_{k=j_0}^{j} \Theta_{A,j,k} = \mathcal{C}_\Phi(w_j + A^{(j)}) - \mathcal{C}_\Phi(w_j + A^{(\ell)}). \tag{4.4}$$

The second endpoint belongs to the accepted finite-low object.

Every a_k, $k \le j$, is $\mathcal{F}_{j-1}$-measurable, so its Malliavin derivative in a fresh root direction v vanishes. Differentiating (4.3) gives the complete atom

$$\boxed{\begin{aligned} T_{A,j,k}[v] = \{&D\Phi(B_{j,k} + a_k) - D\Phi(B_{j,k})\}[v]^T DB_{j,k} \\ &+ D\Phi(B_{j,k} + a_k)[v]^T Da_k \\ &+ \{\Phi(B_{j,k} + a_k) - \Phi(B_{j,k})\}^T Dv. \end{aligned}} \tag{4.5}$$

Equivalently,

$$\begin{aligned} T_{A,j,k}[v] = \int_0^1 \{&D^2\Phi(B_{j,k} + ra_k)[a_k, v]^T D(B_{j,k} + ra_k) \\ &+ D\Phi(B_{j,k} + ra_k)[v]^T Da_k \\ &+ D\Phi(B_{j,k} + ra_k)[a_k]^T Dv\}\, dr. \end{aligned} \tag{4.6}$$

Thus

$$\sum_{k=j_0}^{j} T_{A,j,k} = \mathbb{D}_j\{\mathcal{C}_\Phi(w_j + A^{(j)}) - \mathcal{C}_\Phi(w_j + A^{(\ell)})\}. \tag{4.7}$$

All three R-084 derivative channels are retained. The kth coefficient path contains only control shells below or equal to k; future control shells are absent.

R-081 gives $\|D\Phi\|_\infty \le 3$, while the fixed floor and target heat give $\|D^2\Phi\|_\infty \le M_{e,2}$. Therefore

$$\boxed{|T_{A,j,k}[v]| \le M_{e,2}|a_k||v||DB_{j,k}| + 3|v||Da_k| + 3|a_k||Dv|.} \tag{4.8}$$

If the root basis obeys

$$\sum_h |v_h|^2 \le \kappa_0 2^{-j}, \qquad \sum_h |Dv_h|^2 \le \kappa_1 2^j, \tag{4.9}$$

then the elementary three-term square inequality yields

$$\sum_h |T_{A,j,k}[v_h]|^2 \le C_e\{2^{-j}|a_k|^2|DB_{j,k}|^2 + 2^{-j}|Da_k|^2 + 2^j|a_k|^2\}. \tag{4.10}$$

Equations (4.8)–(4.10) remove the artificial complete translated Z^6 multiplier from the algebraic endpoint. They do not prove far-frequency decay. The two-point Nemytskii secants in (4.5) still require a production paracomposition estimate, and distinct k atoms remain nonorthogonal after squaring.

For the exact sequential atoms, set $v_{A,j,k,t} = P_t^{(j)} T_{A,j,k}$. The exact missing bridge is: for some $s > 0$,

$$\sum_A \int_0^\infty e^{-2t}\mathbb{E}\|\Pi_m P_t^{(j)} T_{A,j,k}\|_{\mathrm{HS}}^2 dt \le 2^{-2s(m-k)}\widetilde{q}_k, \qquad m \ge j + C_0, \tag{4.11}$$

together with

$$\sum_k \widetilde{q}_k \lesssim 1 + \mathbb{E}[X^{1/2}Y^{1/2}]. \tag{4.12}$$

Theorem 3.1 would then close the exact Cartan OU target. A direct integrated CFAR estimate, summing moderate harmonics before taking an extreme-tail bound, would also suffice; a per-atom ledger is a sufficient architecture, not a logical necessity.

5 Quartic Besov one-use payload

Lemma 5.1

Let $0 < s < 1$ and $A \in H^2(\mathbb{T}_L^3) \cap L^6(\mathbb{T}_L^3)$. Then

$$\boxed{\|A^{\otimes 3} \otimes DA\|_{B^s_{1,1}} \le C_{s,L} \|A\|_{H^2} \|A\|_6^3.} \tag{5.1}$$

The statement applies componentwise to the production vector field.

Proof. Use a smooth inhomogeneous Littlewood–Paley partition and assign ties by a fixed ordering. If the differentiated factor has largest frequency j, Holder gives

$$\|\Delta_j DA\, A^3\|_1 \le C2^j \|\Delta_j A\|_2 \|A^3\|_2 \le C2^j \|\Delta_j A\|_2 \|A\|_6^3. \tag{5.2}$$

If an undifferentiated factor $\Delta_j A$ has largest frequency, two other undifferentiated factors are placed in L^6, and Bernstein gives

$$\|DS_{j+c}A\|_6 \le C2^j \|A\|_6. \tag{5.3}$$

Consequently

$$\|\Delta_j A\, A\, A\, DS_{j+c}A\|_1 \le C2^j \|\Delta_j A\|_2 \|A\|_6^3. \tag{5.4}$$

Let P_j denote any product family whose largest input scale is j. Although its output may lie arbitrarily below j through high–high-to-low interactions, it always has output scales $n \le j + c$. The uniform L^1 multiplier bound and $s > 0$ give the geometric low-output summation

$$\sum_{n \le j+c} 2^{sn} \|\Delta_n P_j\|_1 \le C_s 2^{sj} \|P_j\|_1. \tag{5.5}$$

Combining (5.2)–(5.5) over the finitely many largest-input product families reduces the Besov norm to

$$\sum_j 2^{(s+1)j} \|\Delta_j A\|_2. \tag{5.6}$$

Finally,

$$\begin{aligned} \sum_j 2^{(s+1)j} \|\Delta_j A\|_2 &= \sum_j 2^{-(1-s)j} \{2^{2j} \|\Delta_j A\|_2\} \\ &\le \left(\sum_j 2^{-2(1-s)j} \right)^{1/2} \|A\|_{H^2}. \end{aligned} \tag{5.7}$$

The final series converges exactly when $s < 1$; the low-output sum used $s > 0$. This proves (5.1), including high–high-to-low outputs. □

With

$$X_A = \|A\|_{H^2}^2, \qquad Y_A = \|A\|_6^6, \tag{5.8}$$

the right side of (5.1) is $X_A^{1/2} Y_A^{1/2}$: one use of each critical control budget. Equivalently, for $R \in B^{-s}_{\infty,\infty}$,

$$|\langle R, A^{\otimes 3} \otimes DA \rangle| \le C_s \|R\|_{B^{-s}_{\infty,\infty}} X_A^{1/2} Y_A^{1/2}. \tag{5.9}$$

The terminal variational sextic is

$$Y = \|U + A\|_6^6, \tag{5.10}$$

not Y_A. The triangle inequality gives

$$\|A\|_6^3 \le C\{\|U + A\|_6^3 + \|U\|_6^3\}. \tag{5.11}$$

On the regular orthogonal no-revisit one-shot shell class, the canonical K_k smoothing and R-079 (6.4) give the pathwise square-function comparison $X_A = \|A\|_{H^2}^2 \le CX$. Only on this explicitly stated class, (5.11) yields

$$\mathbb{E}[X_A^{1/2} Y_A^{1/2}] \le C\mathbb{E}[X^{1/2}Y^{1/2}] + C(\mathbb{E}X)^{1/2}(\mathbb{E}\|U\|_6^6)^{1/2}. \tag{5.12}$$

The last term is bounded by

$$\delta \mathbb{E}X + C_{\delta,L} \tag{5.13}$$

using the canonical sixth moment. This is form-payable with an arbitrarily small energy allocation, but it is not literally the right side of (4.12) without another bridge. No such pathwise comparison is asserted for a general progressive revisiting control.

6 Corrected translated-model obstruction

R-087's extracted majorant has the form

$$q_k^{\mathrm{mod}} = C_e 2^{-(6\alpha-1)k} \sup_{j\ge k} \sum_A \mathbb{E} \int_0^1 Z_{j,r}^6 \{\|a_k\|_2^2 + \|Da_k\|_2^2\}\, dr. \tag{6.1}$$

The direct-root theorem asks only for $\sum_k q_k^{\mathrm{mod}}$. Nevertheless, the R-087 predictable rare branch still prevents this extraction. Let $N = 2^k$, let $E \in \mathcal{F}_{k-1}$ have probability $p = N^{-6}$, and take

$$h_k = N^3 \mathbf{1}_E e_N, \qquad a_k = K_k h_k = N\mathbf{1}_E e_N. \tag{6.2}$$

Then

$$\mathbb{E}X \asymp \mathbb{E}Y \asymp \mathbb{E}\sqrt{XY} \asymp 1, \tag{6.3}$$

but, with $\beta = 6\alpha - 1$,

$$q_k^{\mathrm{mod}} \asymp N^{-\beta} p N^{6(1+\alpha)} N^4 = N^5 = p^{-5/6}. \tag{6.4}$$

The stronger weighted ledger grows as $N^6 = p^{-1}$. The normalization repair saves one spatial power but does not rescue this majorant.

This fixture does not evaluate the exact complete production atom, so it is not a counterexample to that atom. For bookkeeping only, impose the *unproved toy ansatz* that its one-shell contribution has size

$$pN^2 A^{2+2s}. \tag{6.5}$$

Under $pN^4A^2 \le 1$ and $pA^6 \le 1$, arithmetic within that ansatz gives its maximum at $A = N$, namely N^{-2+2s}; at $s = 7/12$ this is $N^{-5/6}$. Neither the sequential identity nor the executables derive the ansatz (6.5), because its factor $2^{-s(m-k)}$ is precisely the open far-frequency theorem. The value $N^{-5/6}$ is therefore a conditional diagnostic and is not evidence for the production atom. The only rigorous conclusion here is that the N^5 fixture refutes the extracted Z^6 majorant route, not the exact atom.

7 Rational completion as a conditional packet

Let

$$\begin{gathered} A_\eta = B_1 + 2\eta I, \qquad K_\eta = LA_\eta^{-1}L, \qquad M_\eta = L - K_\eta, \\ Q = GG^T - \Gamma, \qquad D_\eta = \frac{1}{2}K_\eta : \Gamma, \end{gathered} \tag{7.1}$$

and

$$\mathcal{P} = \frac{1}{2}L : Q + G^T L c + \frac{1}{2}c^T B_1 c. \tag{7.2}$$

R-087 proved

$$\mathcal{P} + \eta|c|^2 = \frac{1}{2}(c + A_\eta^{-1}LG)^T A_\eta (c + A_\eta^{-1}LG) + \frac{1}{2}M_\eta : Q - D_\eta. \tag{7.3}$$

Pointwise null identity

The apparent debt is not an independent loss:

$$\boxed{\frac{1}{2}G^T K_\eta G - \frac{1}{2}K_\eta : Q - D_\eta = 0.} \tag{7.4}$$

Indeed, $K_\eta : Q = G^T K_\eta G - K_\eta : \Gamma$. Hence the debt is exactly paired with the K_η part of the retained square and transformed Wick term. Paying it after deleting its companion can manufacture a cutoff-divergent loss.

Theorem 7.1 (conditional moment criterion)

Let $\eta > 0$. On a fixed finite-dimensional real space suppose that B_1 and Γ are symmetric positive semidefinite, L is symmetric, $A_\eta = B_1 + 2\eta I$ is positive definite, and B_1, L, c, Γ are measurable with respect to a sigma-field $\mathcal{H}$. Assume that G has a finite conditional second moment and that every displayed product is integrable. Put

$$\mu = \mathbb{E}[G \mid \mathcal{H}], \qquad V = \operatorname{Cov}(G \mid \mathcal{H}). \tag{7.5}$$

Then

$$\boxed{\begin{aligned}\mathbb{E}[\mathcal{P} + \eta|c|^2 \mid \mathcal{H}] = {} & \frac{1}{2}(c + A_\eta^{-1}L\mu)^T A_\eta (c + A_\eta^{-1}L\mu) \\ & + \frac{1}{2}M_\eta : \mu\mu^T + \frac{1}{2}L : (V - \Gamma).\end{aligned}} \tag{7.6}$$

Conditional Gaussianity is unnecessary.

Proof. Take conditional expectation in (7.3). Since

$$\mathbb{E}[GG^T \mid \mathcal{H}] = V + \mu\mu^T, \tag{7.7}$$

the square contributes its mean square plus $K_\eta : V/2$. Combining that covariance term with $M_\eta : V/2$ gives $L : V/2$, while $-(K_\eta + M_\eta) : \Gamma/2 = -L : \Gamma/2$. The mean terms combine as shown in (7.6). ☐

If $\mu = 0$ and $V = \Gamma$, then

$$\mathbb{E}[\mathcal{P} \mid \mathcal{H}] = \frac{1}{2}c^T B_1 c \geq 0. \tag{7.8}$$

The retained-square variance cancels D_η exactly. This recovers the fresh predictable covariance-matched branch already closed more generally by the complete R-077 Doob cancellation. It does not close the surviving coefficient-dominant packet, whose coefficient and carrier may share a probability root.

A nonzero conditional mean leaves $M_\eta : \mu\mu^T/2$, and covariance mismatch leaves $L : (V - \Gamma)/2$. Neither defect is repaired by changing η. In the scalar fixtures

$$\eta = 1, \quad B_1 = 0, \quad L = -1, \quad V = \Gamma = 1, \quad \mu = t, \quad c = t/2, \tag{7.9}$$

formula (7.6) equals $-3t^2/4$. With $\mu = c = 0$, $L = -1$, $V = 2$, and $\Gamma = 1$, it equals $-1/2$.

Standalone-debt and adapted-alignment no-go

For a production-shaped cutoff diagnostic keep the target dimension fixed, say $d = 6$, and take

$$B_1 = 0, \quad L = \ell I_6, \quad c = 0, \quad \Gamma_N = \rho_N I_6, \qquad \mathbb{E}G = 0, \quad \operatorname{Cov}(G) = \Gamma_N, \qquad \rho_N \uparrow \infty. \tag{7.10}$$

Then $\mathbb{E}\mathcal{P} = 0$, while

$$D_{\eta,N} = \frac{\ell^2}{4\eta}\operatorname{Tr}\Gamma_N = \frac{6\rho_N \ell^2}{4\eta}. \tag{7.11}$$

The expected completed-square variance is exactly the same quantity. Therefore standalone payment can diverge with cutoff covariance at fixed target dimension even though the original packet has zero mean.

The same-root alignment

$$B_1 = 0, \quad L = 2\eta, \quad \Gamma = 1, \quad c = -G \tag{7.12}$$

has $M_\eta = 0$, zero completed square, and

$$\mathcal{P} + \eta c^2 = -\eta \tag{7.13}$$

pathwise. This is a method no-go, not a production counterexample: c is not predictable relative to G. Together (7.11)–(7.13) fire `NG-2026-07-25-A13-RATIONAL-STANDALONE-ETA-DEBT-AND-K-HEAT`.

8 Matrix-fractional heat boundary

Let P be an averaging operator, let A be positive definite and L symmetric, and write $\bar{A} = PA$, $\bar{L} = PL$. Direct expansion gives

$$\boxed{\begin{aligned} &P(LA^{-1}L) - \bar{L}\bar{A}^{-1}\bar{L} \\ &\quad = P\Big[(L - \bar{L}\bar{A}^{-1}A)A^{-1}(L - A\bar{A}^{-1}\bar{L})\Big] \succeq 0. \end{aligned}} \tag{8.1}$$

The second factor is the transpose of the first. Thus $A \mapsto LA^{-1}L$ has a positive matrix-fractional Jensen defect.

This positivity does not give a sign after Wick contraction. If $\mathcal{J}$ denotes the left side of (8.1), then $\mathcal{J} : Q$ may have either sign; for example $G = 0$ and $\Gamma = I$ give $Q = -I$ and $\mathcal{J} : Q = -\operatorname{Tr}\mathcal{J}$. Therefore K_η does not inherit the native linear backward-heat telescope of B_1 in a form that can be separated from Q.

The genuine open rational object is

$$\boxed{\text{endpoint square} + \frac{1}{2}M_\eta : Q - D_\eta + \text{backward-heat transport} + \text{complete lower-chaos forest.}} \tag{8.2}$$

These pieces must remain together. The complete fresh-root forest cancels under R-077; the same-root adapted defect in (8.2) does not.

Correct completion-parameter budget

The pinned control coefficient is

$$\epsilon_v = \frac{9}{20}, \qquad q = \frac{1}{2\epsilon_v} = \frac{10}{9}. \tag{8.3}$$

If η is reallocated inside this 9/20, q remains 10/9. The number

$$\frac{1}{2p} - \epsilon_v = \frac{5}{11} - \frac{9}{20} = \frac{1}{220}, \qquad p = \frac{11}{10}, \tag{8.4}$$

is only the strict comparison margin when η is added on top of 9/20. In that added mode,

$$q_\eta = \frac{10}{9 + 20\eta}, \qquad q_\eta - p = \frac{1 - 220\eta}{90 + 200\eta}, \tag{8.5}$$

so strict comparison requires $\eta < 1/220$. No universal 1/220 cap applies to an internal allocation.

9 Attempt and evidence map

Route	Verdict	Exact consequence
R-084 normalization audit	corrected attribution	The exact OU target has no outer 2^j.
Direct-root Schur	proved	$s > 0$, unweighted $\sum_k q_k$, constant (3.6).
R-085 weighted Schur	retained	Valid for its stronger 2^j-weighted expression.
Sequential shell telescope	exact	Three channels retained; no future control shell in the kth path.
Sequential tame envelope	proved	Removes the generic total-Z^6 algebraic multiplier.
Sequential far atom	open	Requires the production two-point estimate (4.11).
Quartic Besov payload	proved	$0 < s < 1$ and pure-control $X_A^{1/2} Y_A^{1/2}$.
Terminal sextic bridge	form-payable	Adds $\delta \mathbb{E} X + C_{\delta,L}$, not literally (4.12).
Old q_k^{mod} extraction	failed method	Unweighted growth N^5; weighted growth N^6.
Toy secant ansatz	conditional diagnostic only	$N^{-5/6}$ follows only after assuming the unproved far-decay ansatz (6.5).
Fresh-root or Wick alone	insufficient	Conditional zero chaos and same-root coefficient defects survive.
Localization or BMO	failed method	Expected budgets do not yield predictable essential-supremum control.
Standalone trace payment	failed method	Creates $(\ell^2/(4\eta)) \operatorname{Tr} \Gamma_N$ while the original mean can be zero.
Conditional rational theorem	exact criterion	Closes only centered covariance-matched predictable blocks.
Matrix-fractional Jensen	PSD but signed after Q	No separate backward-heat telescope for $K_\eta : Q$.

The minimal remaining dependency chain is

$$\begin{aligned} &\text{sequential Cartan two-point bridge or direct integrated CFAR} \\ &+ \text{complete rational same-root causal square/trace/forest bound} \\ &\Longrightarrow \text{REG} \\ &\Longrightarrow \text{uniform OVERLAP on the R-087 variational core} \\ &\Longrightarrow \text{R-066 one-use synthesis} \Longrightarrow q = 10/9 \text{ Nelson.} \end{aligned} \tag{9.1}$$

R-080 supplies the low object, R-083 the polynomial CFAR and full Pauli–Fierz forest, R-084 the linear rows, R-086 the paid rational branches, and R-087 the fixed-cutoff variational CORE. No all-progressive packet theorem is needed between OVERLAP and that CORE invocation.

10 Devil's-advocate review

1. **"The direct-root theorem dropped a required 2^j." DISMISSED within the stated target.** Equation (2.2) is the exact R-084 expression and has no such factor. The R-085 theorem remains valid for (2.4). Any rescaling must be exhibited explicitly.
2. **"The weaker threshold proves Cartan CFAR." UPHELD as an overclaim.** Theorem 3.1 only performs the triangular summation. The production bridge (4.11)–(4.12), or a direct integrated substitute, remains open.
3. **"R-087's old q_k^{mod} now satisfies the ledger." DISMISSED.** The rare branch gives $q_k^{\mathrm{mod}} \asymp N^5$. The repair saves one power but does not make the extracted majorant bounded.

4. **“The rare branch refutes the exact Cartan atom.” DISMISSED.** It evaluates the generic Z^6 majorant, not the exact atom. The $N^{-5/6}$ arithmetic uses the explicitly unproved toy ansatz (6.5) and cannot be used as positive evidence either.

5. **“Sequential telescoping makes control shells orthogonal.” DISMISSED.** The telescope occurs before the Hilbert–Schmidt square. Cross-k terms remain and are handled only conditionally by Theorem 3.1.

6. **“The quartic Besov lemma proves the literal terminal ledger.” VALID WITH MITIGATION.** It proves the pure-control critical product. Passing to $U + A$ adds (5.11)–(5.13) on the stated one-shot class, which is form-payable but not literally (4.12).

7. **“The trace debt can be paid separately by Young.” DISMISSED.** Equation (7.4) and the fixed-target-dimension fixture (7.10) show that the debt is the variance part of the retained square. Separating it can create a cutoff-covariance-divergent loss from a zero-mean packet.

8. **“Conditional centering closes the rational remainder.” UPHELD as an overclaim.** The centered covariance-matched branch was already closed by R-077. The coefficient-dominant residual can have nonzero conditional mean, covariance defect, and same-root lower chaoses.

9. **“Positive Jensen defect gives a favorable Wick sign.” DISMISSED.** A PSD matrix contracted with $Q = GG^T - \Gamma$ has no fixed sign.

10. **“The value $1/220$ caps every completion parameter.” DISMISSED.** It caps only an additional charge if strict $p < q_\eta$ is required. An internal share of the pinned $9/20$ leaves $q = 10/9$ unchanged.

11 Executable verification and next theorem

The primary executable verifies the two predecessor normalizations directly from their notes, all three geometric factors, aligned finite Schur fixtures, the sharp $s = 0$ boundary, symbolic sequential and radial secant identities, non-tautological tame-current fixtures, root Hilbert–Schmidt bookkeeping, quartic dyadic summation, the old-majorant exponents and conditional toy-ansatz arithmetic, rational completion and conditional identities, standalone-debt and adapted fixtures, matrix-fractional Jensen, signed Wick contraction, and the η budget.

The non-importing audit independently uses seeded vector Schur tests, direct finite differences of a nonlinear current, shell telescopes, dyadic Cauchy tests, random positive matrix conditional moments, pointwise trace null tests, random matrix-fractional Jensen tests, and every no-go fixture. It imports no primary module.

Run from the repository root:

```
python codes/foundations/
  a13_classii_direct_root_cartan_schur_sequential_secant_
  rational_conditional_trace_reduction_verify.py
```

The next analytic theorem should keep expectation inside the exact sequential coefficients and prove (4.11)–(4.12), or prove the direct integrated CFAR target without a per-atom ledger. In parallel, it should bound the complete object (8.2), not its trace debt in isolation. Only after both are available can REG and uniform OVERLAP be assembled.

Result footer

Result ID:	A13-CLASSII-DIRECT-ROOT-CARTAN-SCHUR-SEQUENTIAL-SECANT-RATIONAL-CONDITIONAL-TRACE-REDUCTION
Ledger:	R-088.
Precise statement:	The exact unweighted R-084 OU target has a nonorthogonal Schur bound for every s>0. At s=7/12 its balanced constant is 16.30295538482827... and its gap is 2^(-7C/6). Sequential physical-shell secants give an exact three-channel Cartan atom and tame root envelope. A quartic Besov lemma pays the pure-control X_A^(1/2)Y_A^(1/2) payload for 0<s<1. The rational K_eta square, K_eta:Q Wick term, and eta debt obey an exact three-term null identity; centered matched square variance cancels the debt conditionally. A conditional theorem isolates mean/covariance defects, and matrix-fractional Jensen does not give a Wick sign.
Scope:	Finite cutoff and fixed positive floor, under the hash-pinned R-084--R-087 conventions. The analytic Besov lemma is deterministic. Rational conditional statements assume the displayed measurability and finite conditional second moments.
Dependencies:	R-050; R-066; R-074; R-077; R-079; R-081; R-083--R-087.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/a13_classii_direct_root_cartan_schur_sequential_secant_rational_conditional_trace_reduction_verify.py
Expected output:	Primary, non-importing independent, integrated, and aggregate manifest-pinned PASS; exit zero.
Audit:	AUDIT-2026-07-25-A13-R085-CARTAN-OUTER-WEIGHT-NORMALIZATION.
Negative result:	NG-2026-07-25-A13-RATIONAL-STANDALONE-ETA-DEBT-AND-K-HEAT (method no-go, not production counterexample).
Falsification gate:	Failure of a geometric factor, secant identity, Besov exponent, rational conditional identity, Jensen certificate, evidence hash, PDF, or assertion contract.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No production sequential secant-to-quartic bridge, direct integrated Cartan CFAR, coefficient-dominant rational causal packet, REG, uniform OVERLAP, controlled-shell one-use, Nelson, measure theorem, removal limit, or Sector-A closure.
Next required action:	Prove the expectation-inside sequential Cartan bridge or direct integrated CFAR and the complete rational same-root square/trace/forest causal packet; then reassemble REG and prove uniform OVERLAP.